ModelIt!

Student Activity Pack

Why Do Objects Move Differently?

Balanced and Unbalanced Forces

Grade 3 | NGSS 3-PS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think it is easier to push a shopping cart on a smooth floor than on grass?

2 What happens when you push a ball on ice versus on carpet? Which goes farther and why?

3 Draw a picture showing all the forces acting on a ball when you kick it across a field.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Push Force | Friction | Object Motion

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gentle Push on Smooth Floor

Push Force set to low, surface is smooth

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Hard Push on Rough Carpet

Push Force set to high, surface is rough

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Hard Push on Smooth Floor

Push Force set to high, surface is smooth

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Object Weight — How heavy the object is — heavier objects are harder to push because they press harder on the surface, creating more friction

Surface Type — What the surface is made of — smooth surfaces like ice create less friction than rough surfaces like sandpaper

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does a ball roll when you push it, but a book on a table stays perfectly still? Use evidence from your simulation to support your answers.

1 Why does a book sitting on a table not fall through the table or float up into the air?

2 How did your model help you understand why objects move differently on different surfaces?

8 STEM Challenge: Design the Ultimate Sliding Race

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YOUR ENGINEERING MISSION

Design a race course that tests how different surfaces and push forces affect how far objects slide, proving that friction and force determine motion.

THE SCENARIO

A toy company wants to create a game where players push objects across different surfaces to see which goes the farthest. Your team must design and test a race course that uses at least three different surfaces to show how friction affects motion!

Engineering Design Thinking

1. WHAT THREE SURFACES WILL YOU TEST — TILE, CARPET, SANDPAPER, WOOD, ICE?

2. HOW WILL YOU MAKE SURE YOU PUSH WITH THE SAME FORCE EACH TIME — USING A RAMP OR A RULER LAUNCHER?

3. WHAT OBJECTS WILL YOU SLIDE — A BLOCK, A TOY CAR, A COIN?

4. HOW WILL YOU MEASURE HOW FAR THE OBJECT TRAVELS ON EACH SURFACE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Mechanical Engineers study forces and motion to design machines, vehicles, and equipment that move efficiently and safely. They earn $60,000-$100,000/year.